



ScholarNest

Spark Azure Databricks

Databricks Spark Certification and beyond

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Azure
Databricks



Absolute Beginner to Specialization in Apache Spark and Azure Databricks

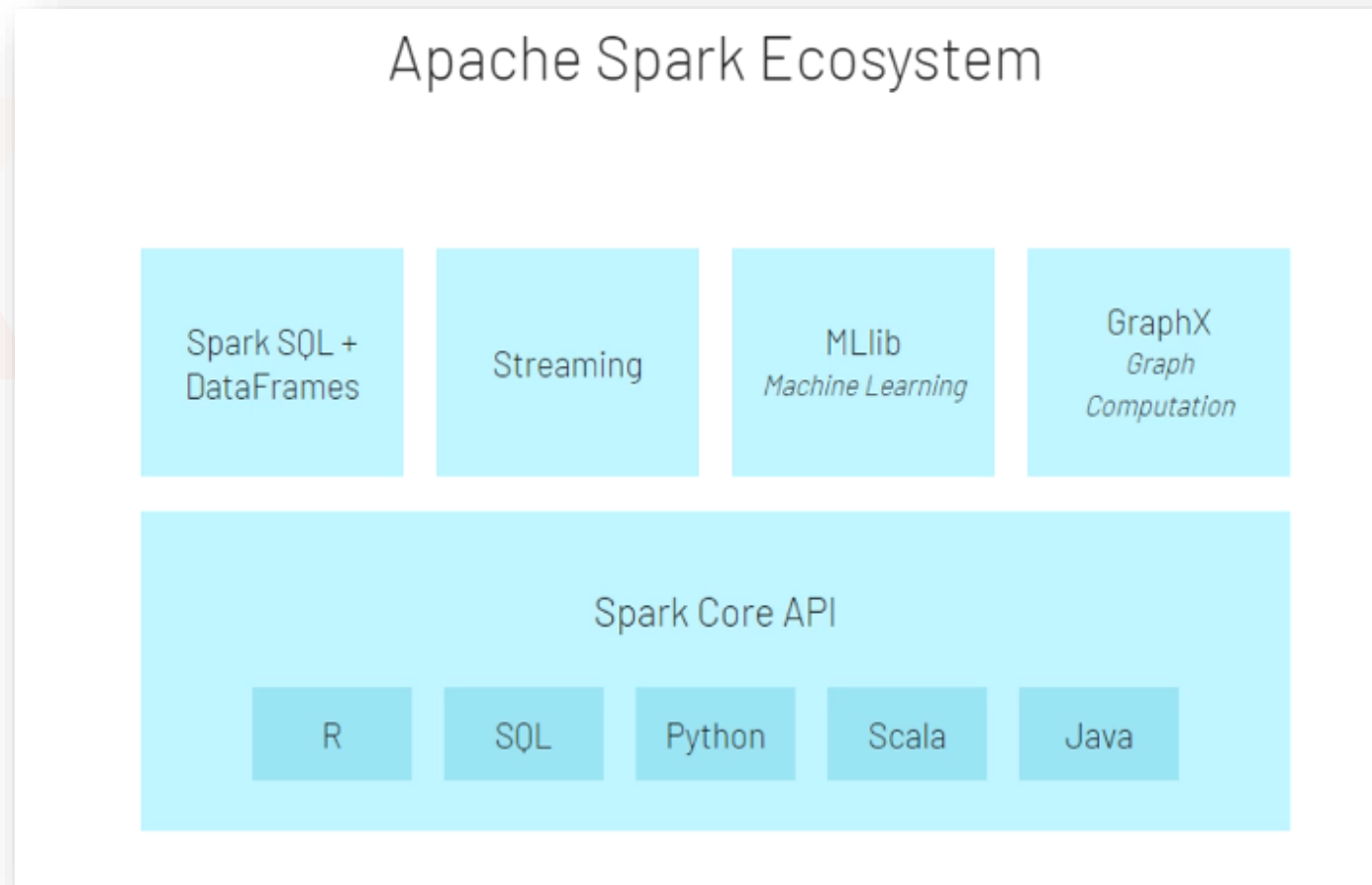


Apache Spark and Databricks Cloud - Introduction

Apache Spark is a distributed data processing framework, and this diagram represents the Spark Ecosystem.

The Spark ecosystem is designed in two layers:

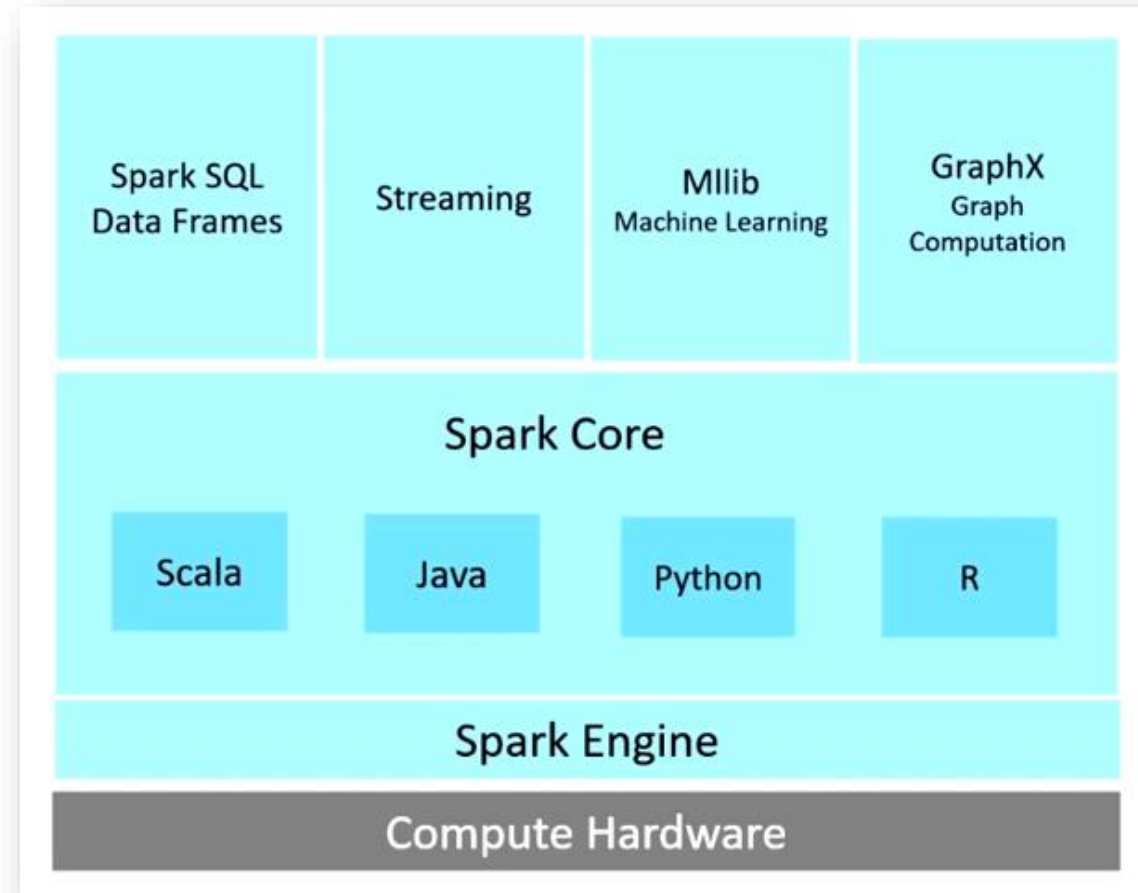
1. The top layer is a set of DSLs, libraries, and APIs.
2. The bottom layer is the Spark Core Layer.



The Spark core layer itself has got two parts:

1. A distributed Computing Engine
2. A set of Core APIs

And this whole thing runs on a cluster of computers to offer you distributed data processing. However, Spark does not manage the cluster. It only gives you a data processing framework.

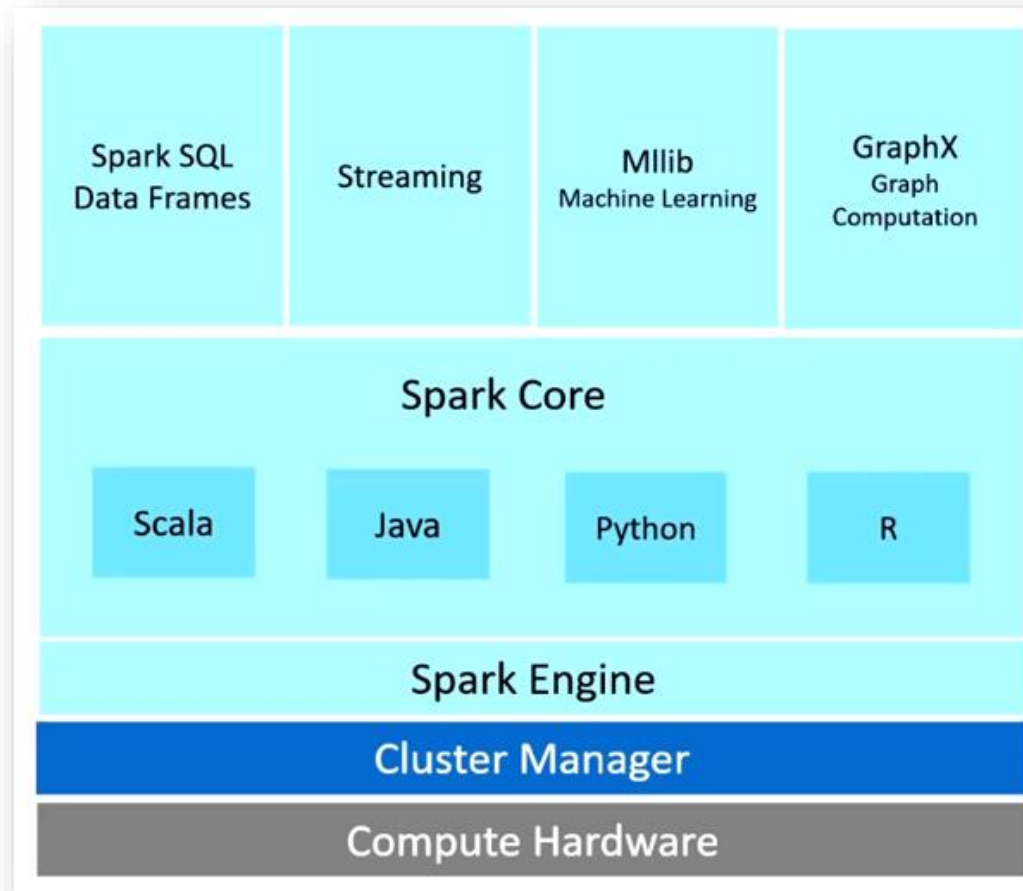


The Spark core layer itself has got two parts:

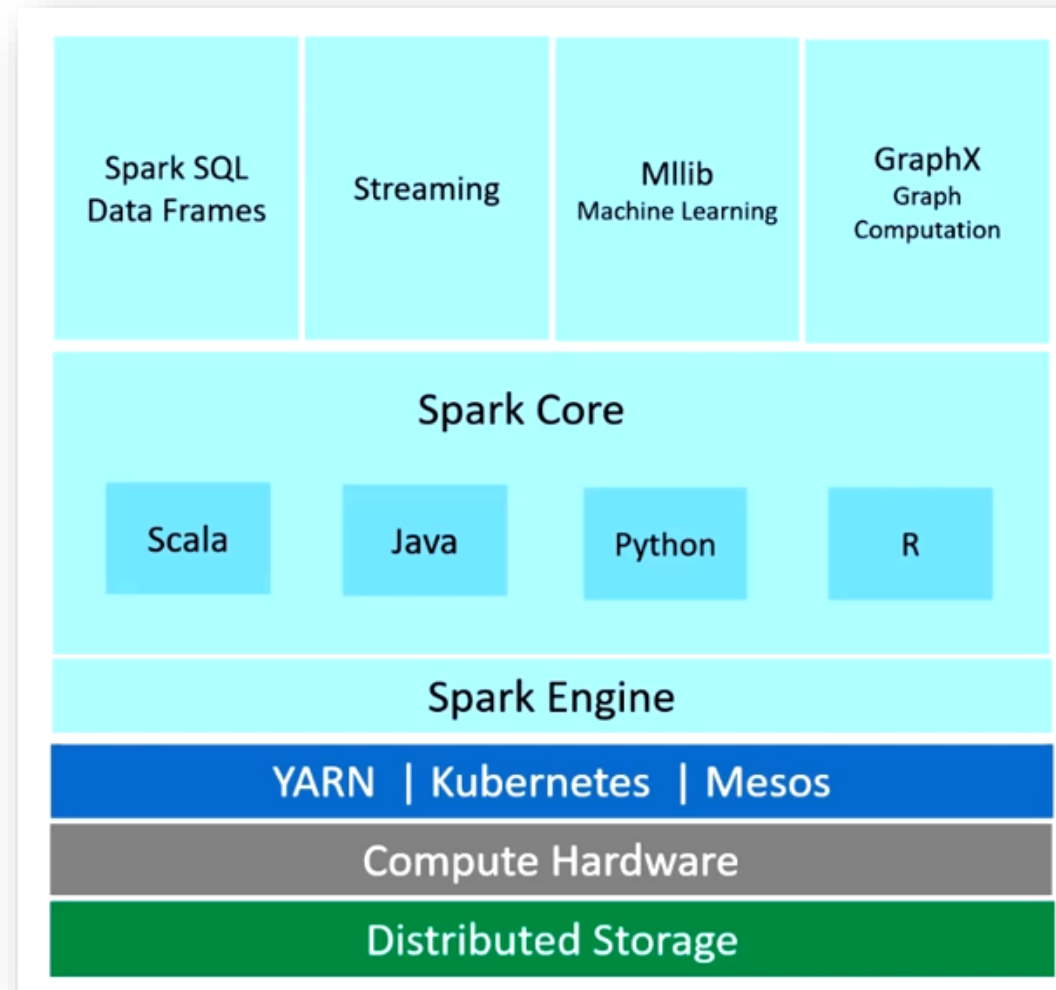
1. A distributed Computing Engine
2. A set of Core APIs

And this whole thing runs on a cluster of computers to offer you distributed data processing.

However, Spark does not manage the cluster. It only gives you a data processing framework. Some examples of Cluster Manager are: YARN, Mesos, Kubernetes



Spark also doesn't come with an in-built storage system. And it allows you to process the data, which is stored in a variety of storage systems. The most popular and commonly used storage systems are HDFS, Amazon S3, Azure Data Lake Storage, Google Cloud Storage, and the Cassandra file system.



Apache Spark does not offer Cluster Management and Storage Management.

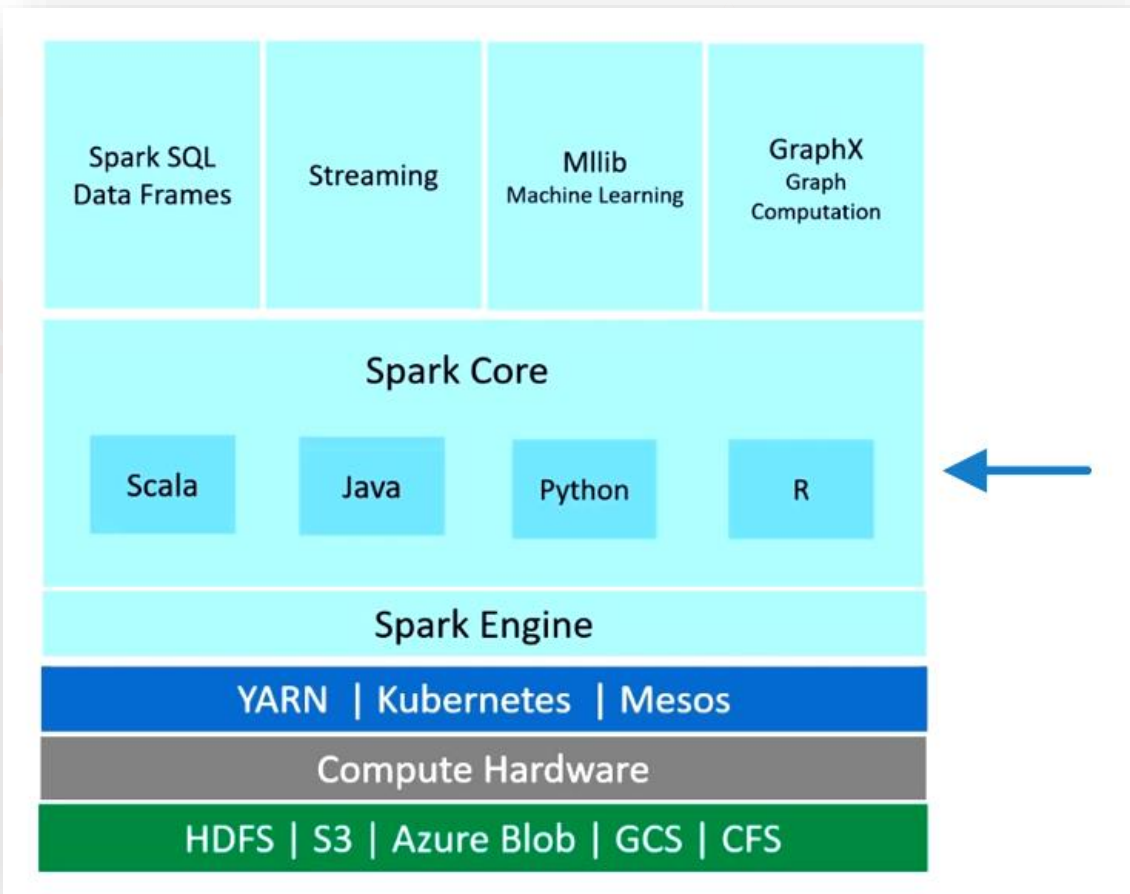
All you can do with Apache Spark is to run your data processing workload.

And that part is managed by the Spark Compute Engine.

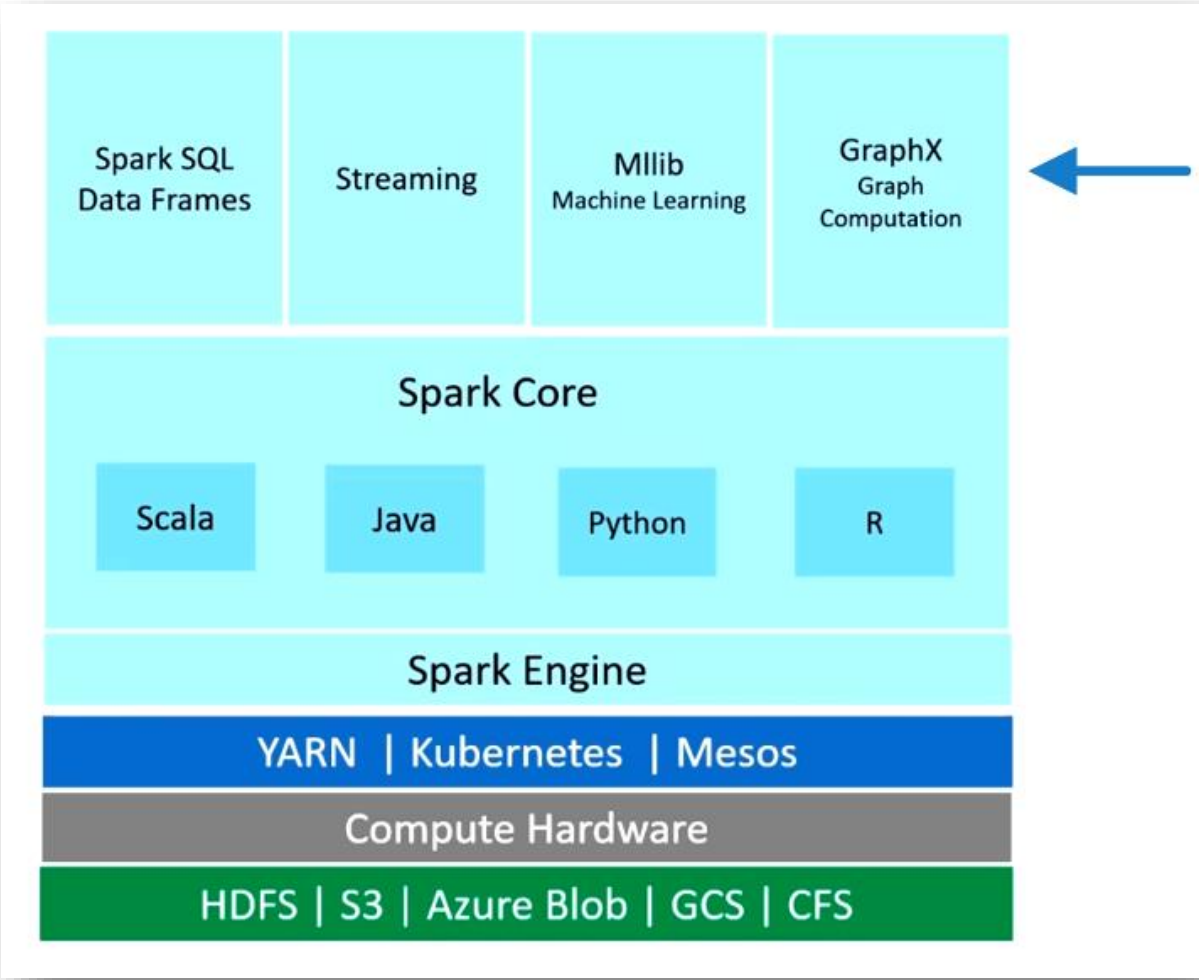
So the Spark compute engine is the core that runs and manages your data processing work and provides you with a seamless experience.

All you need to do is submit your data processing jobs to Spark, and the Spark core will take care of everything else.

The Core API layer is the programming interface layer that offers you the core APIs in four major languages: Scala, Java, Python, and R. These are the APIs that we used to write data processing logic during the initial days of Apache Spark. However, these APIs were based on resilient distributed datasets (RDD). These APIs are the most complicated way to work with Apache Spark and they also lack some performance optimization features. However, these APIs can offer you the highest level of flexibility to solve some complex data processing problems.



The topmost layer is the prime area of interest for most Spark developers and data scientists. It is a set of libraries, packages, APIs, and DSL. These are developed by the Spark community over and above the Core APIs. The topmost API layer is grouped into four categories to support four different data processing requirements. However, this is just a logical grouping, and there is no rigid boundary.



Let us look at the four categories of the top most API layer:

1. Spark SQL and DataFrame/Dataset APIs - Spark SQL allows you to use SQL queries to process your data. Both of these together can help you resolve most of the structured and semi-structured data crunching problems.
2. Spark Streaming libraries - They allow you to process a continuous and unbounded stream of data.
3. A set of libraries specifically designed to meet your machine learning, deep learning, and AI requirements.
4. Graph Processing libraries, and they allow you to implement Graph Processing Algorithms using Apache Spark.

So the topmost layer is nothing but a set of libraries and DSLs to help you solve your data crunching problems.

And all these are available in multiple languages such as Java, Scala, Python, and R.

There are three reasons why Spark Ecosystem is so popular:

1. Abstract - Spark will abstract away that you are coding to execute your program on a cluster of computers. So, all the complexities of distributed storage, computation, and parallel programming, is abstracted away by the Spark core.
2. Unified Platform - Spark combines the capability of SQL queries, Batch Processing, Stream Processing, Structured and semi-structured data handling, Graph processing, machine learning, and deep learning. All of this in a single framework using your favourite programming language. You can mix and match them to solve many sophisticated requirements.
3. Ease of Use - Comparing it with old Hadoop and MapReduce code, Spark code is much shorter, simpler, and easy to read and understand.

Now let's come to Databricks. Apache Spark is an open-source project. The original creators of Apache Spark donated it to Apache Foundation and made it an Open Source project. However, that same team formed a company and a commercial product around Apache Spark. The company and the product are both named Databricks. And Databricks offers the following things over and above Apache Spark:

1. Spark on Cloud Platform
2. Spark Cluster Management
3. Notebooks and Workspace
4. Administration Controls
5. Optimized Spark
6. Databases/Tables and Catalog
7. Databricks SQL Analytics
8. Delta Lake Integration
9. ML Flow
10. Industry vertical solutions and accelerators



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